

YUVRAJ SINGH

singhy5@myumanitoba.ca | linkedin.com/in/yuvixmahar | github.com/yuvixmahar | yuvixmahar.github.io

EDUCATION

University of Manitoba, Winnipeg, MB · B.Sc. in Computer Engineering (Co-op)

Expected Apr 2028

EXPERIENCE

Controls Engineer Intern · Winnipeg School Division

Jun 2026 – Aug 2026

- Restored full 4-stage cooling on a rooftop unit previously capped at 1–2 of 4 stages by rewriting a faulty GCL+ staging threshold and coordinating contractor commissioning.
- Diagnosed control issues across 10 AHUs/RTUs and 4 HRVs/ERVs in 5 school buildings via live BACnet tracking in Delta Controls enteliWEB, working with the district's Energy Management Coordinator.
- Cataloged 8 sensor and logic defects across HVAC control programs in a formal issues report, driving corrective code fixes and enteliWEB graphic redesigns.

Software Engineer Intern · Quark Power Inc.

Sep 2025 – May 2026

- Shipped a full-stack license management platform (React/TypeScript, FastAPI, MySQL, Redis) as the sole engineer, exposing 60+ REST endpoints for provisioning, authentication, machine registration, and telemetry.
- Owned the on-premise production deployment on Ubuntu Linux, configuring Nginx and systemd for long-running services, with 8 MySQL tables versioned through Alembic migrations.
- Secured the API with JWT authentication and per-user/per-machine Redis rate limiting, and moved blocking email workflows to asynchronous background tasks.

Embedded Systems Member · University of Manitoba Robotics Team

Oct 2024 – Present

- Built an ESP32 gripper controller for a 6-DoF robotic arm that filters CAN frames by message ID, decodes the commanded angle, and drives a servo in real time; integrated a third-party CAN library after vetting it against the transceiver.
- Wrote a ROS2 C++ publisher node for a BNO055 IMU over I2C that publishes timestamped orientation, gyroscope, and acceleration data for sensor fusion and navigation.

PROJECTS

Wireless Environmental Monitor | Raspberry Pi Pico 2 W, MicroPython, I2C

May 2026

- Built a standalone sensor node on a Raspberry Pi Pico 2 W (RP2350, Arm Cortex-M33) that reads a DHT11 temperature/humidity sensor and shows live readings on a 16×2 LCD over I2C.
- Ran an HTTP server on the microcontroller over Wi-Fi, sharing a single event loop with sensor polling and LCD updates, to serve a live dashboard to any browser on the network with no host machine.

BISONplan | bisonplan.vercel.app | Python, FastAPI, React, Docker, GitHub Actions

May 2026 – Present

- Launched a course-planning platform for University of Manitoba students that reached 450+ users, 2,000+ page views, and 50k+ API requests in its first month.
- Proxied the university's Aurora (Banner SSB) registration system live instead of mirroring stale data, falling back to age-stamped cached results with freshness banners during upstream outages.
- Guarded schedule correctness with interval-overlap and date-range conflict detection, backed by 170+ automated tests across a 6-job CI pipeline.

JobRadar | github.com/yuvixmahar/job-radar | Python, asyncio, httpx, SQLite, Pydantic

Jul 2026 – Present

- Built a job-posting monitor with one adapter per ATS platform, not per company (8 integrations incl. Workday, Greenhouse, Lever), loaded as entry-point plugins behind JobSource/Notifier interfaces.
- Fetches all sources concurrently with asyncio under a semaphore cap so one failing source can't break a run, and deduplicated postings with a SQLite primary key and one batched INSERT OR IGNORE per poll.
- Maintained 95% coverage across 230 tests under strict mypy with fully mocked HTTP; boundary-aware matching keeps C, C++, C#, and .NET distinct.

SKILLS

Languages: C, C++, Python, TypeScript, JavaScript, SQL, Bash, Java

Embedded: ESP32, STM32 (HAL), RP2350 (Pico 2 W), Arduino, MicroPython, CAN bus, I2C, UART, SPI, PWM, ROS2

Backend & Web: FastAPI, REST, JWT, SQLAlchemy, Alembic, asyncio, httpx, Pydantic, React, Vite, Tailwind CSS

Data & Infrastructure: MySQL, PostgreSQL, SQLite, Redis, Docker, Linux (Ubuntu), Nginx, systemd, GitHub Actions, Cloudflare

Controls & Tools: BACnet, Delta Controls enteliWEB, GCL+, Git, pytest, Vitest, mypy; currently learning NumPy and pandas